Training motor control

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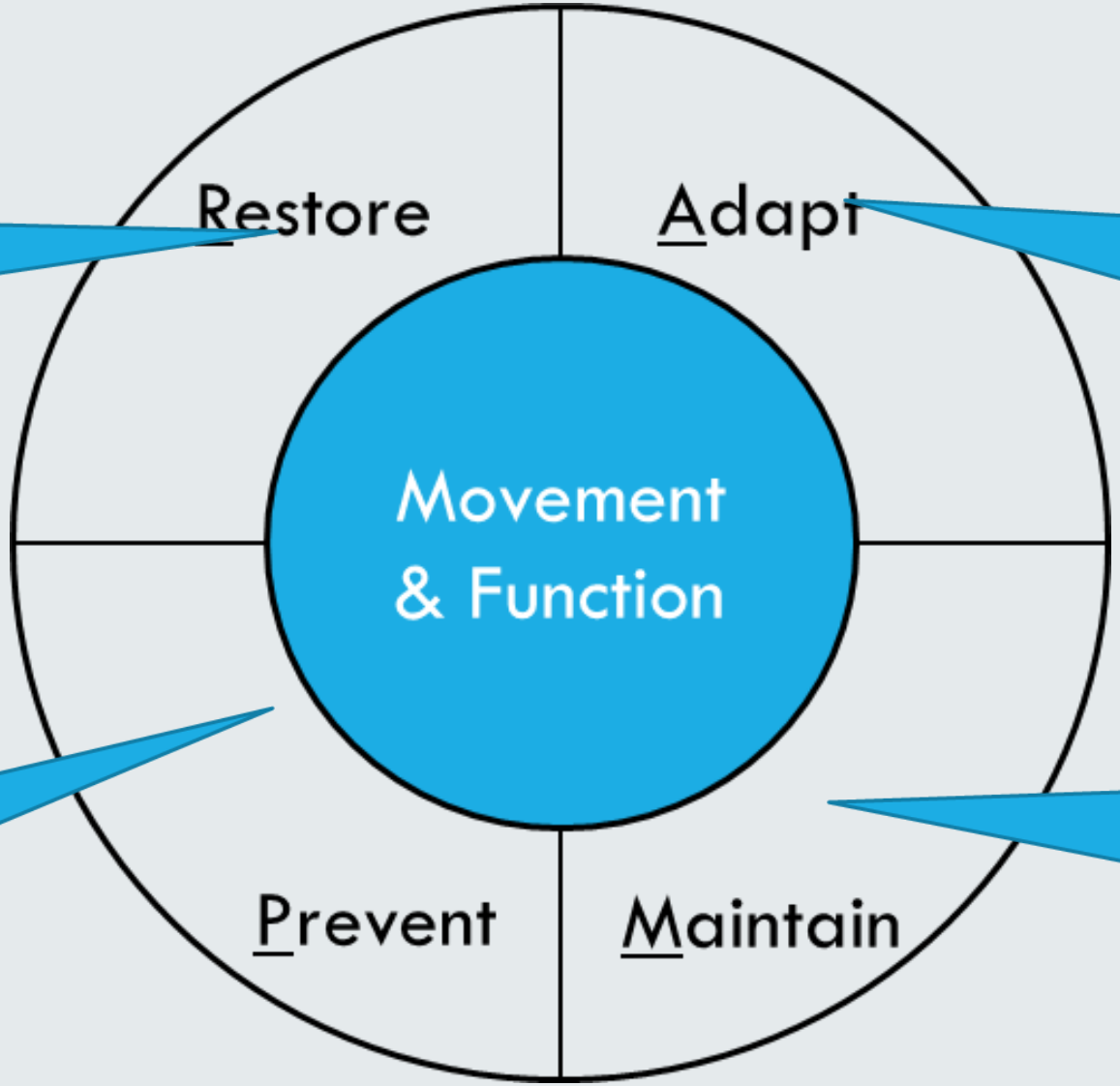
Learning Outcomes

By the end of this lecture, you should be able to:

- Explain the conceptual framework used in neurological Physiotherapy
- Identify and demonstrate core components of a motor training programme.
- Highlight treatment approaches used in neurorehabilitation

Role of the Physiotherapist in contemporary neurological Practice

The restoration of movement and function to pre-impairment status



The use of alternative movement strategies to complete a task

Prevention of complications

Maintenance of function especially in progressive conditions

What are the most appropriate outcomes for evaluating the effect of intervention

What is the most appropriate way to examine my patient

What is the best approach to intervention?

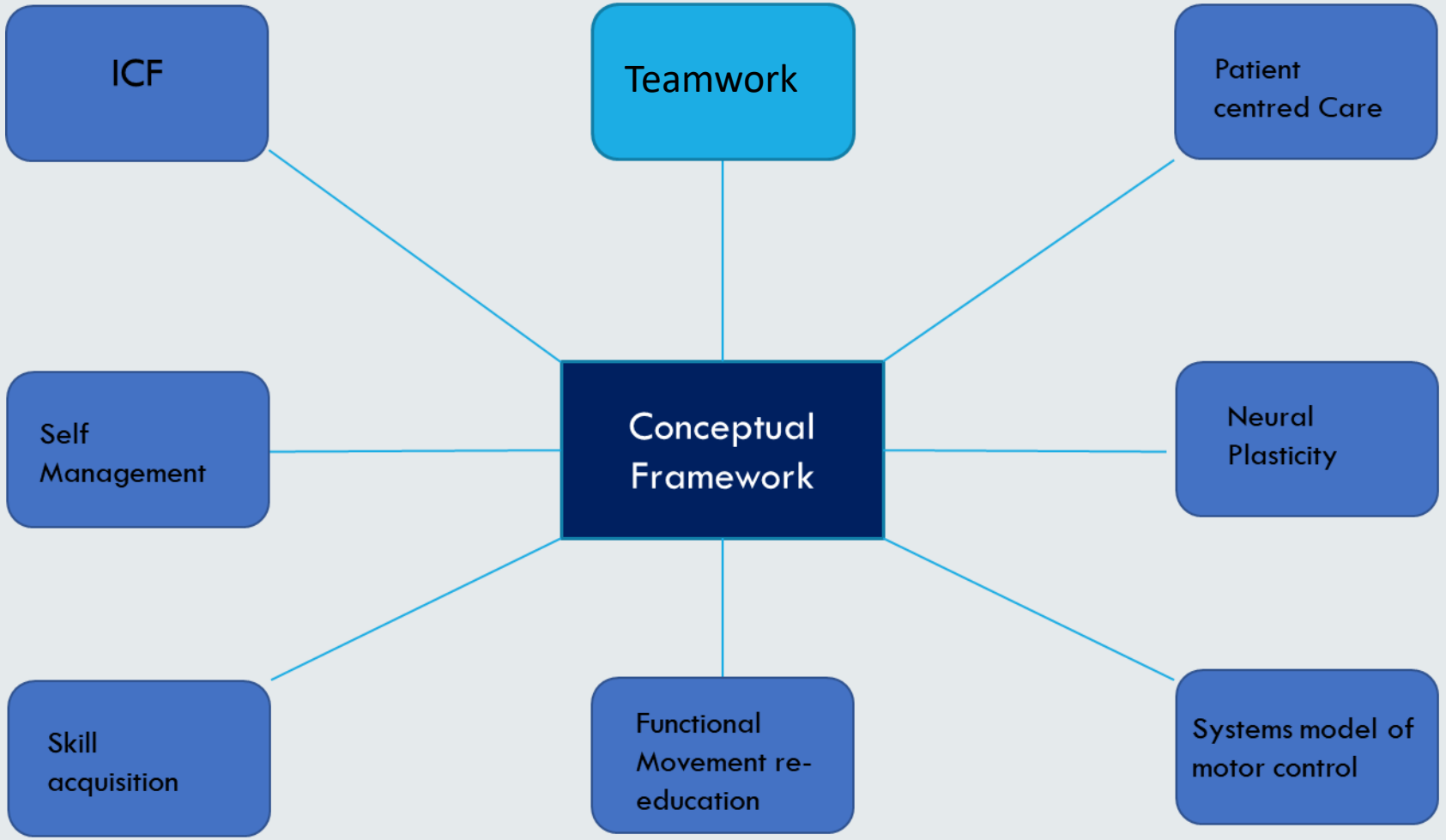


What criteria should be used to decide what the priority problems are?

How do I establish goals that are realistic and meaningful?

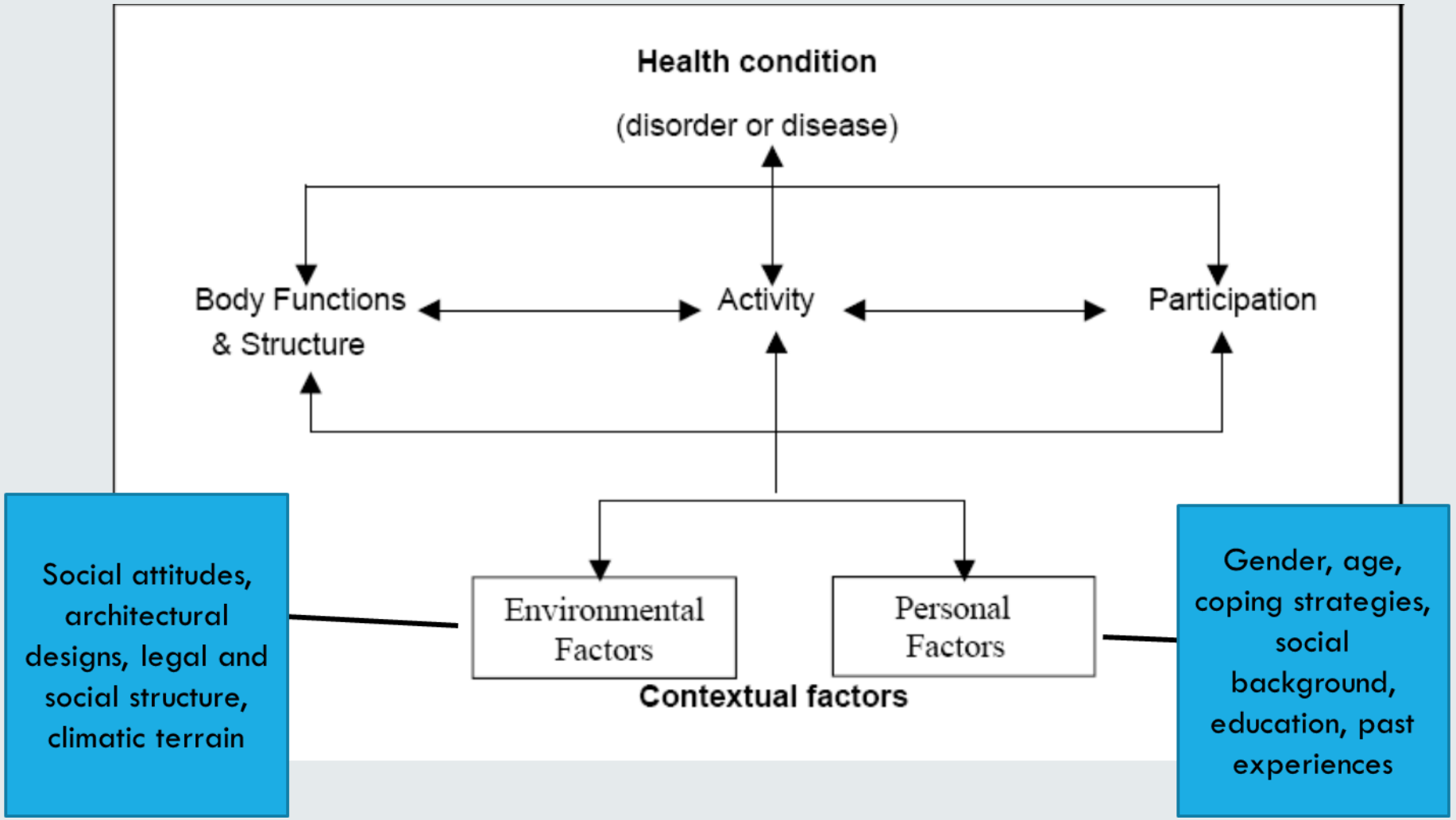
How much time should be spent on documenting functional ability vs. evaluating underlying impairments?

Components of a conceptual framework for Neurological Rehabilitation



Lennon & Bassile (2009)

International Classification of Functioning (ICF, 2001)



TEAMWORK

Teamwork approach is required

Multiple domains of health are affected

Multi-disciplinary team
- Co-ordinate rehabilitation
- Agree common plan of care
- Goal setting
- Plan discharge and long-term destination



Patient centred care



"Shared"

- Control
- Decision about interventions
- Management priorities/goals
- Focus on the whole patient
- Psychosocial context

Neural plasticity

Neural plasticity: capacity of the brain to alter it function and structure in response to injury, use and non-use

Spontaneous recovery occurs after neurological damage; there is also neural plasticity

- Enduring changes in neural function and structure following damage, experience and therapy.

The brain responds to injury by adaptation aimed at restoring function

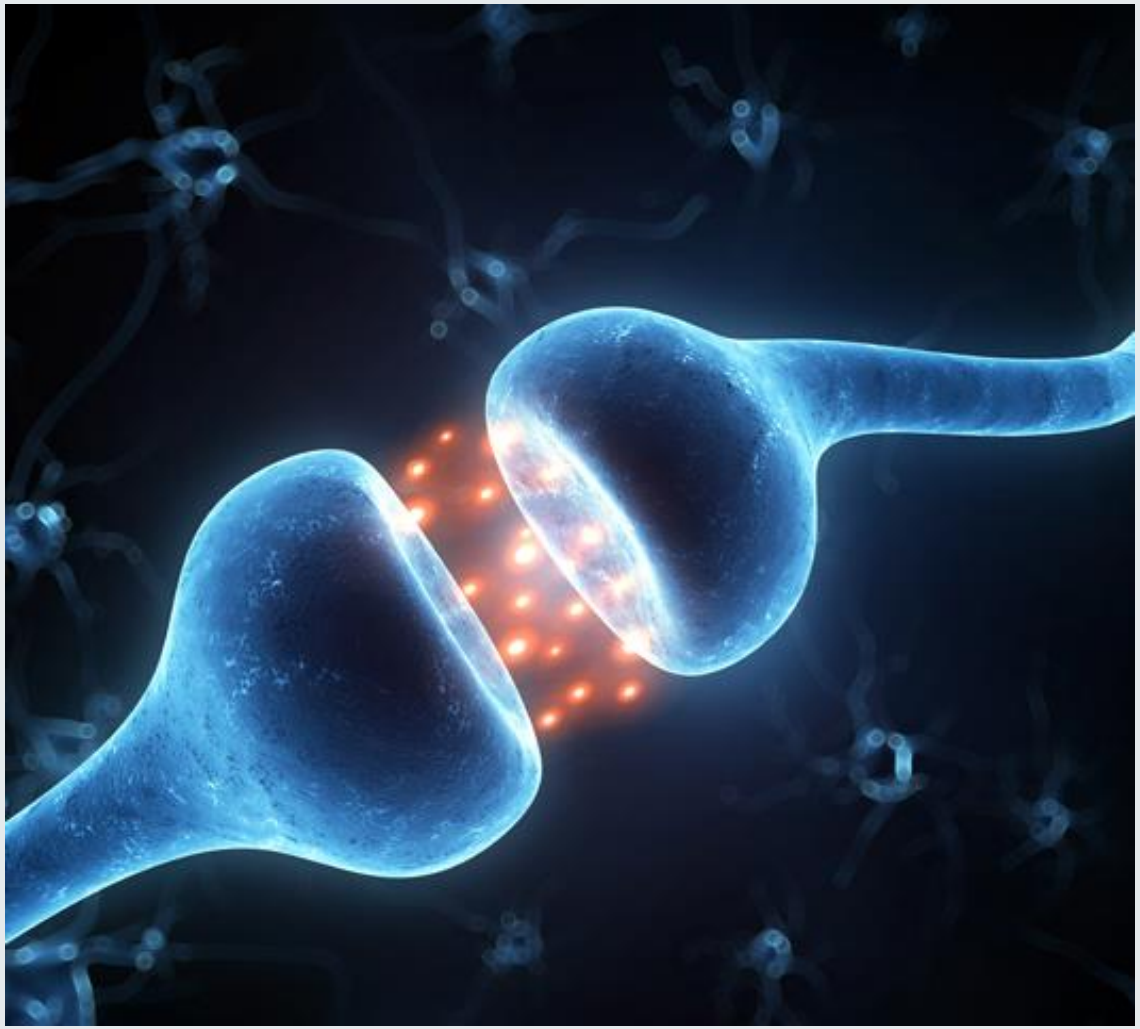
- Cortical map enlargement to adjacent or remote regions of the brain
- Cortical maps can also be modified by inputs such as sensory, experience and learning.

This implies that therapy could be directed towards the enhancement of these processes for recovery of movement and function especially at the early phase of recovery.

Techniques to promote neuroplasticity:

- Afferent information, task specific exercises etc

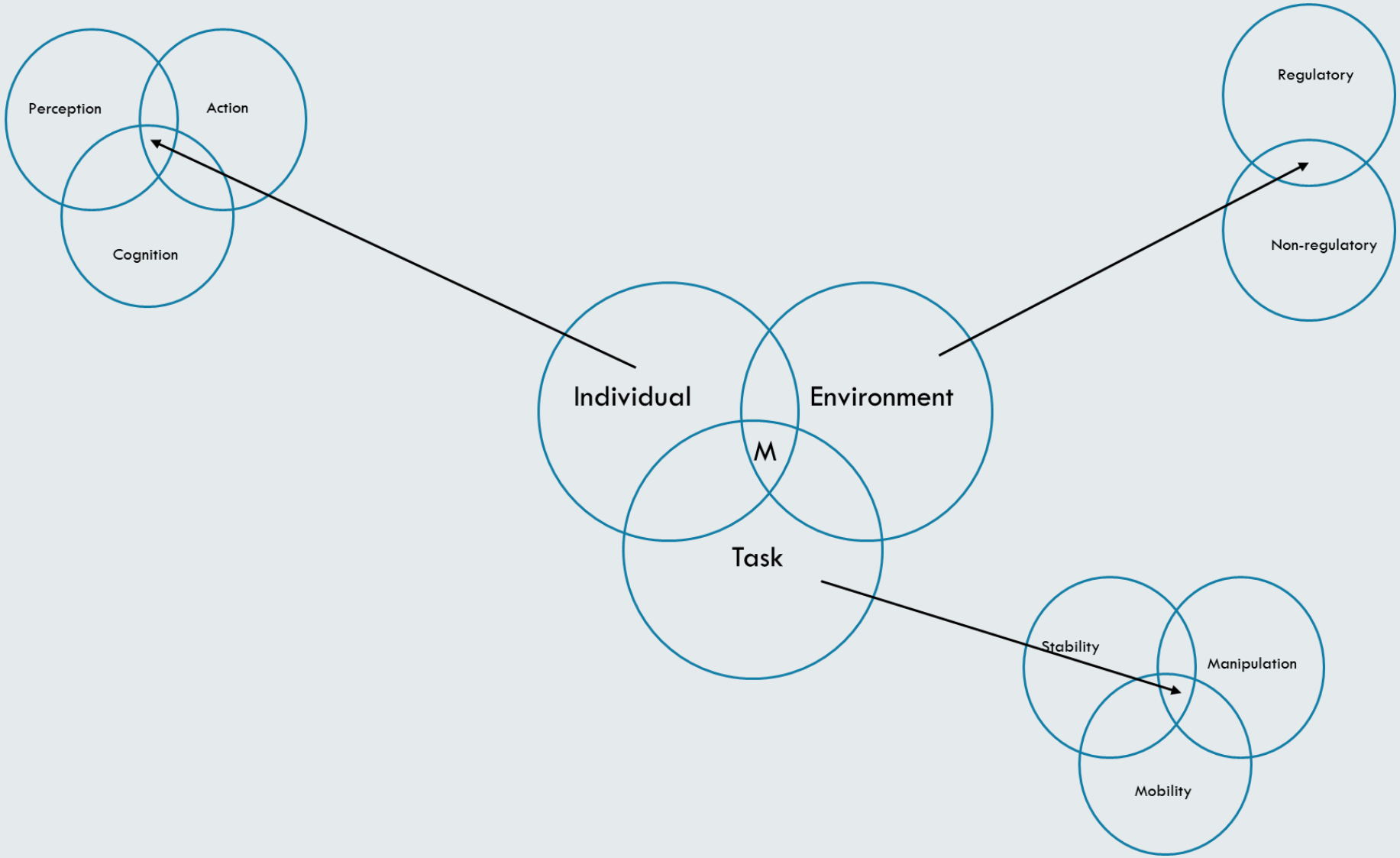
Later stage rehabilitation would centre of approaches that alter non-use of arm



Systems model of motor control

- Models of motor control – Reflex, systems, hierarchical, ecological etc
- Recommended model of motor control is the Systems model (Shumway Cook & Woollacott)
- Integrates evidence from neurophysiology, biomechanics and motor learning
- Suggests that control of integrated movement is distributed throughout many interacting systems.
- What are the implication of a systems model for therapy planning?

Implication for treatment planning



Functional Movement Re-education

Key components

- Understand what normal movement is:
  - Walking – indoors, outdoors
  - Reaching and grasp
  - Transfers – mobility in bed, lying to sitting, sit to stand, stairs etc

- Emphasis is placed on training control of muscles and promoting learning across relevant tasks
  - Therapeutic handling
  - Practice of functional tasks

Motor Learning/Skills Acquisition

- Task specific training is required for functional recovery in neurological patients

- Evidence from motor learning & skills training can be utilised in structuring therapy sessions.

Key strategies

- Practice – amount of practice, practice conditions, mental practice
- Feedback - intrinsic, extrinsic – knowledge of results, knowledge of performance.
- Transfer to everyday activities – practice activities outside therapy

Self-Management

- An essential part of neurological rehabilitation – to promote behaviour change.

“ an individual’s ability to manage the symptoms, treatment, physical and psychosocial consequences and lifestyle changes inherent in living with a chronic condition”

Involves:

- Ability to monitor one’s condition
- Affect the cognitive, behavioural and emotional responses necessary to maintain a satisfactory quality of life
- Establishing a dynamic and continuous process of self-regulation

Evidence based Rehabilitation

- **Evidence based Medicine as defined by Sackett et al. 1996**

"is the conscientious, explicit, and judicious use of current best evidence in making decisions about the care of individual patients"

Philosophical approach that integrates:

- Best available research, Clinical reasoning and expert opinion

For EBP, physiotherapists are expected to use a wide range of strategies supported by current evidence base

- Cochrane reviews, other meta-analysis or high-quality systematic reviews
- Multi-disciplinary clinical guidelines e.g. Stroke, MS, PD etc

- There are still so many key areas with no evidence or conflicting evidence
- Clinical reasoning skills
- Patient & carers' needs, wishes and goals

PRINCIPLES OF TRAINING MOTOR CONTROL

1. Optimise muscle functioning including:
   - Soft tissue flexibility
   - Muscle strength
   - Muscle timing
   - Physical Fitness

2. Improve neuromuscular control of body segments and skills in everyday activities using task-oriented practice

Muscle strengthening in neurology

1. Repetitive contraction of a weak muscle group using single or multi-joint weightlifting exercise, elastic band, isometric or isokinetic exercise etc.

2. Functional strength training – strengthen force generation, train inter-segmental control of the limb and preserve muscle length e.g. close chain type step-up, leg press, squatting, sit-to-stand, treadmill walking.

3. Practice of functional actions made ineffective by muscle weakness and reduced limb control e.g. stair climbing & descent, ramp walking, overground and treadmill walking with the aim for increasing speed and distance walked. Repetitive standing and sitting down holding a tray, reaching for, picking up and manipulating objects on the ground and above head.

Notes

- The first type is aimed at individuals with paralysed or denervated muscles.
- Second and third types are aimed at people with ability to contract relevant muscles but with insufficient force, poor timing of force, poor control of limb or lack of agility.
- Training parameters – up to 10 reps with no rest; 3 sets with short breaks
- Exercises should be progressed.

Promoting Physical fitness in neurology

- Cardiorespiratory endurance is tested with the 6 minutes' walk test

Inpatient and outpatient exercise therapy or "rehabilitation"		
Aerobic		
• Large-muscle activities (eg, walking, graded walking, stationary cycle ergometry, arm ergometry, arm-leg ergometry, functional activities seated exercises, if appropriate	• Increase walking speed and efficiency • Improve exercise tolerance (functional capacity) • Increase independence in ADLs • Reduce motor impairment and improve cognition • Improve vascular health and induce other cardioprotective benefits (eg, vasomotor reactivity, decrease risk factor)	• 40%–70% $\dot{V}O_2$ reserve or HR reserve; 55%–80% HR max; RPE 11–14 (6–20 scale) • 3–5 d/wk • 20–60 min/session (or multiple 10-min sessions) • 5–10 min of warm-up and cool-down activities • Complement with pedometers to increase lifestyle physical activity

Taken from Ballinger SA et al. Stroke 2014 45(8) 2532-53

Improving neuromuscular control using Task Specific Exercise

- Improving performance should be specific to what should be achieved.
- Mechanisms underlying training motor control: neuroplasticity & Motor Learning

Implementing Neuroplasticity motor control programmes	Applying motor learning principles to motor control programmes
Meaningful activities (Salience) e.g. sit to stand to get to toilet, to get food, to go out	**High Intensity Practice:** Within and outside practice
High Intensity: 3 – 4 times a week, at least 2 hours per day crucial	**Feedback:** intrinsic & extrinsic feedback
High repetition: 300 reps for upper limb and 2000 steps for lower limb	**Practice conditions:** modification of practice elements to enable task to be performed
Task specificity: acquisition of skills required and not anatomical movements	**Focus of attention:** Paying attention to interaction with objects rather than movements is more meaningful
Challenge: Task must involve dual tasking and/or multi-tasking	**Instructions:** modelling or verbal instructions
Variation and Timing: arrange task to start from various starting position	

Specific Treatment techniques in neurological rehabilitation - facilitatory techniques

Tapping



https://youtu.be/4b2UiVTlNLw

Brushing



https://youtu.be/0B23ngwLcGc

Specific Treatment techniques in neurological rehabilitation

Brief Icing



https://youtu.be/nE7HcZqMEgs

Passive stretching - quick



https://youtu.be/tRlpD_cQtzQ

Specific treatment techniques in neurological rehabilitation

Joint compression



https://youtu.be/2acoY3HAskc

Muscle vibration



https://youtu.be/9Tmd7Umn1Qg?t=16

Specific treatment techniques: normalisation of tone & soft tissue length

Passive stretching - slow



https://youtu.be/YaxyWv1wZvM?t=91

Night splints



Specific treatment techniques: normalisation of tone & soft tissue length

Weight bearing



Used to reduce contractures

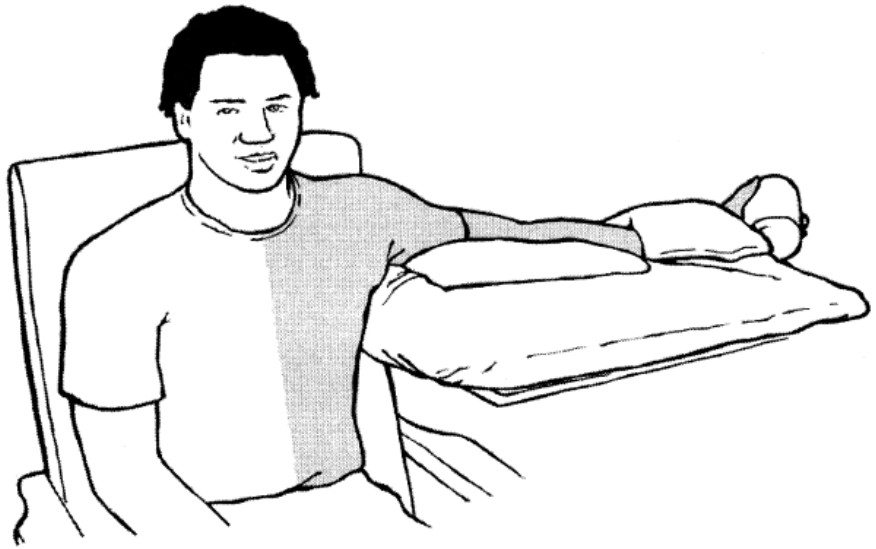
Serial casting



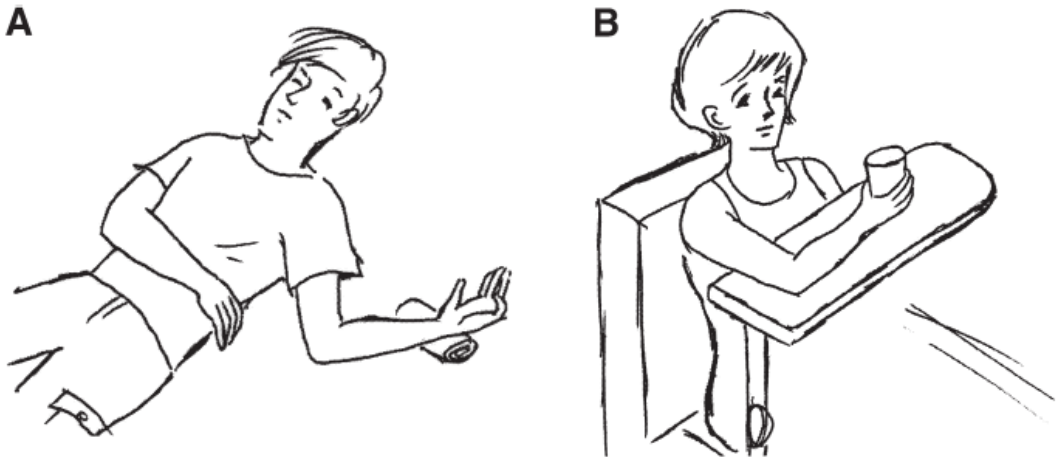
Used to prevent or reduce contractures

Specific treatment techniques: normalisation of tone & soft tissue length

Positioning



Positioning



Ada et al., 2005

Other approaches are

Technique	Uses
Pressure	Facilitate and inhibit a response in muscles
Prolonged icing	Reduces afferent and effect neurotransmission
Exercise approaches	
Hydrotherapy	Improved ability to move in water
Gymnastic balls	To support patient, train balance etc
Proprioceptive Neuromuscular Facilitation	Promote muscle contraction and reduce tone
Treadmill training	Cardiorespiratory and repetitive mobility training
Pilates	Core , shoulder girdle and limb control
Tai chi	Reducing fear of falling, improve balance
Constrained induced movement training	Forced upper limb use
Cueing	Prompting movement i
TENS	Pain Relief, muscle stimulation, spasticity relief
Neurodynamics	Promoting neural mobility
Orthoses	Promote function and support joints

References

1. Carr J & Shepherd R. Training motor control, increasing strength and fitness and promoting skill. Chapter 2 in Carr J & Shepherd R's Neurological Rehabilitation (2nd edition). Elsevier: Churchill Livingstone. Pages 15 -48.

2. Carr J & Shepherd R. Standing and sitting down. Chapter 4 in Carr J & Shepherd R's Neurological Rehabilitation (2nd edition). Elsevier: Churchill, Livingstone. Pages 77 -94.